

1. Which of the following is an example of an arithmetic operator?
 - A. &&
 - B. +
 - C. >=
 - D. ==
2. What is the main purpose of arithmetic expressions?
 - A. To compare values
 - B. To perform mathematical calculations
 - C. To repeat statements
 - D. To store files
3. Which operator is used for division in most programming languages?
 - A. %
 - B. /
 - C. *
 - D. //
4. What is the result of the expression $10 \% 3$?
 - A. 3
 - B. 1
 - C. 0
 - D. 10
5. Which of the following is a relational operator?
 - A. +
 - B. &&
 - C. <
 - D. =
6. Relational expressions usually return what type of value?
 - A. Integer
 - B. Decimal
 - C. Boolean
 - D. Character
7. Which symbol represents “equal to” in programming?
 - A. =
 - B. ==
 - C. !=
 - D. <=

8. What will the expression `5 > 3` evaluate to?
- A. False
 - B. Error
 - C. True
 - D. Null
9. Which Boolean operator means “AND”?
- A. `||`
 - B. `!`
 - C. `&&`
 - D. `==`
10. What is the result of `TRUE && FALSE`?
- A. TRUE
 - B. FALSE
 - C. NULL
 - D. ERROR
11. Which Boolean operator means “OR”?
- A. `||`
 - B. `&&`
 - C. `!`
 - D. `==`
12. What does the NOT operator (`!`) do?
- A. Adds values
 - B. Multiplies values
 - C. Reverses a Boolean value
 - D. Compares values
13. Which of the following is an example of a Boolean expression?
- A. `5 + 2`
 - B. `x = 10`
 - C. `7 < 9`
 - D. `print("Hello")`
14. What is a control structure in programming?
- A. A type of database
 - B. A mechanism that controls program flow
 - C. A computer virus
 - D. A programming language

15. Which are the three basic control structures?
- A. Input, Process, Output
 - B. Sequence, Selection, Iteration
 - C. Variable, Constant, Operator
 - D. Compile, Execute, Debug
16. Which control structure executes statements one after another?
- A. Selection
 - B. Iteration
 - C. Sequence
 - D. Branching
17. What is the purpose of assignment statements?
- A. To compare values
 - B. To assign values to variables
 - C. To stop loops
 - D. To create functions
18. Which symbol is commonly used for assignment?
- A. ==
 - B. =
 - C. &&
 - D. !=
19. What will happen in the statement `x = 5`?
- A. x is compared to 5
 - B. x is assigned the value 5
 - C. x becomes FALSE
 - D. x is deleted
20. Which of the following is a valid assignment statement?
- A. `10 = x`
 - B. `x == 10`
 - C. `x = 10`
 - D. `x != 10`
21. What is a compound statement?
- A. A single statement only
 - B. Multiple statements grouped together
 - C. A Boolean expression
 - D. A looping condition

22. Which symbols are commonly used to define compound statements in many languages?
- A. []
 - B. <>
 - C. {}
 - D. //
23. Which statement is used to make decisions in programming?
- A. Loop statement
 - B. Selection statement
 - C. Assignment statement
 - D. Arithmetic statement
24. Which keyword is commonly used for branching?
- A. if
 - B. loop
 - C. print
 - D. int
25. What is branching in programming?
- A. Repeating statements continuously
 - B. Choosing different execution paths
 - C. Declaring variables
 - D. Ending the program
26. Which statement is an example of a branching statement?
- A. if-else
 - B. for loop
 - C. while loop
 - D. do-while loop
27. What is the purpose of an if statement?
- A. To repeat instructions
 - B. To execute code based on a condition
 - C. To declare variables
 - D. To create files
28. Which statement executes when the if condition is FALSE?
- A. switch
 - B. else

- C. break
- D. continue

29. Which statement allows multiple choices based on a variable value?

- A. if
- B. switch
- C. while
- D. do-while

30. Which control structure repeats a set of statements?

- A. Selection
- B. Branching
- C. Iteration
- D. Assignment

31. What is another term for iterative statements?

- A. Decision statements
- B. Looping statements
- C. Arithmetic statements
- D. Branch statements

32. Which loop is guaranteed to execute at least once?

- A. for loop
- B. while loop
- C. do-while loop
- D. nested loop

33. Which loop is best used when the number of repetitions is known?

- A. while loop
- B. do-while loop
- C. for loop
- D. infinite loop

34. Which loop continues as long as a condition is TRUE?

- A. while loop
- B. switch statement
- C. if statement
- D. assignment statement

35. What happens if a loop condition never becomes FALSE?

- A. The program stops immediately
- B. Syntax error occurs

- C. Infinite loop occurs
- D. Variable is deleted

36. Which keyword is used to terminate a loop immediately?

- A. continue
- B. break
- C. return
- D. stop

37. What does the continue statement do?

- A. Stops the loop completely
- B. Skips the current iteration and continues the loop
- C. Restarts the program
- D. Deletes the loop variable

38. Which statement combines multiple conditions?

- A. Compound conditional statement
- B. Assignment statement
- C. Arithmetic statement
- D. Output statement

39. What is nested iteration?

- A. A loop inside another loop
- B. A condition inside an assignment
- C. A variable inside a function
- D. A branch inside a comment

40. Which of the following is a selection statement?

- A. if-else
- B. while
- C. for
- D. do-while

41. Which expression evaluates to TRUE?

- A. $10 < 5$
- B. $7 == 7$
- C. $3 > 8$
- D. $2 != 2$

42. What is the result of `!(TRUE)`?

- A. TRUE
- B. FALSE

- C. NULL
- D. ERROR

43. Which operator is used for “not equal to”?
- A. ==
 - B. <=
 - C. !=
 - D. >=
44. Which statement is used to execute one block among many choices?
- A. for
 - B. switch
 - C. while
 - D. continue
45. What type of statement is “ $x = y + 5$ ”?
- A. Selection statement
 - B. Assignment statement
 - C. Iterative statement
 - D. Branching statement
46. Which control structure is represented by “if-else”?
- A. Sequence
 - B. Iteration
 - C. Selection
 - D. Assignment
47. What is the output type of a Boolean expression?
- A. Character
 - B. Decimal
 - C. TRUE or FALSE
 - D. Integer only
48. Which statement helps organize multiple instructions as one block?
- A. Compound statement
 - B. Arithmetic statement
 - C. Relational statement
 - D. Assignment statement
49. Why are control structures important in programming?
- A. They increase file size
 - B. They manage the flow of execution

- C. They replace variables
- D. They prevent calculations

What is the main function of expressions and statements in programming?

- A. To define program logic and operations
- B. To replace hardware
- C. To increase internet speed
- D. To create operating systems